BUSINESS PROPOSAL

ResQ Plus

An AI-Powered Emergency Response & Coordination Platform for Bangladesh

"ResQ Plus doesn't replace emergency services — it's intelligently connects them."

Prepared For	University Idea Contest — Innovation & Entrepreneurship Panel
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Category	Health-Tech / Public Safety / Artificial Intelligence
Date	03 July 2026

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2. Executive Summary

Every year, thousands of lives in Bangladesh are affected not by the absence of hospitals, ambulances, or willing helpers, but by the absence of coordination between them. In the first few minutes of a medical emergency, road accident, fire, or natural disaster, victims and bystanders are often left navigating a fragmented system alone — unsure which number to call, which hospital to go to, or how to keep a family informed while help is on the way. ResQ Plus is built to close that gap.

The Problem

Bangladesh currently lacks a unified, intelligent emergency response system. Citizens rely on scattered phone numbers, informal networks, and guesswork during the most time-critical moments of their lives. Ambulance dispatch is manual and slow, hospital availability is invisible to the public, and family members are often the last to know.

The Solution

ResQ Plus is an AI-powered emergency coordination platform that connects citizens, hospitals, ambulance services, trained volunteers, and emergency contacts through a single intelligent system. Rather than functioning as another isolated SOS button, ResQ Plus acts as an orchestration layer — using artificial intelligence to assess the nature of an emergency, guide the user through immediate first steps, identify the nearest appropriate help, and notify the right people simultaneously.

The Impact

By compressing the decision-making time during the "golden minutes" of an emergency, ResQ Plus has the potential to measurably improve survival outcomes, reduce panic-driven errors, strengthen community-based volunteer response, and give families peace of mind through real-time visibility into a loved one's emergency status.

The Vision

ResQ Plus is designed as the foundation for a national emergency response layer — one that can eventually integrate with government emergency services, hospital networks, and disaster response systems, positioning Bangladesh among countries with digitally coordinated public safety infrastructure.

At a Glance

Sector: Health-Tech × Public Safety × Artificial Intelligence

Core Offering: AI-powered emergency coordination platform connecting citizens, volunteers, hospitals and ambulances

Primary Goal: Reduce emergency response delay through intelligent coordination

Stage: Concept / Prototype-ready startup proposal

3. Problem Statement

Emergency response in Bangladesh suffers not from a single point of failure, but from a chain of small, compounding delays — each one costing precious time when time is the one resource that cannot be recovered. The following are the core problems ResQ Plus is designed to address.

3.1 Delayed Ambulance Response

Ambulance dispatch in most parts of the country remains manual and phone-call dependent, with no centralized system to identify the nearest available vehicle. Callers often contact multiple numbers before reaching a responsive service, losing critical minutes in the process.

3.2 Lack of First-Aid Knowledge

Bystanders present at the scene of an emergency are frequently willing to help but lack the knowledge to act correctly. Incorrect handling of injuries, especially in road accidents, can worsen outcomes rather than improve them.

3.4 Hospital Confusion

Victims and their companions often do not know which nearby hospital is equipped to handle a specific emergency, or whether that hospital currently has bed and specialist availability. This frequently results in patients being transferred between hospitals, losing further time.

3.5 Volunteer Disconnect

Bangladesh has a strong culture of civic volunteerism, particularly visible during road accidents and disasters. However, there is no structured way to alert trained or nearby volunteers the moment an emergency occurs, so this goodwill is rarely converted into fast, organized action.

3.6 Family Notification Delay

Families are typically informed of an emergency only after someone at the scene remembers to call them — often after the situation has already stabilized or worsened. There is no automated, real-time way to keep emergency contacts updated.

3.7 Panic During Emergencies

In high-stress situations, even capable individuals struggle to think clearly. Without guided support, people waste time deciding what to do first, who to call, and where to go — delays that compound every other problem listed above.

> **Why This Costs Lives**
>
> *In medical emergencies, trauma cases, and cardiac events, outcomes are directly correlated with the speed and quality of the first response. Every minute of delay caused by miscommunication, indecision, or lack of coordination is a minute working against survival.*

4. Why This Problem Matters

4.1 The "Golden Minutes" Concept

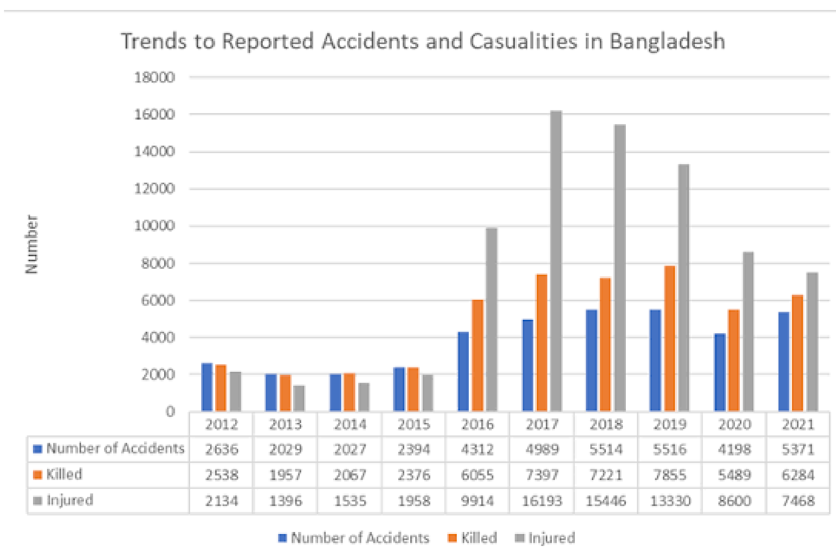
Emergency medicine widely recognizes that the initial minutes following a traumatic injury or acute medical event — often referred to as the "golden hour," and within it, the even more critical "golden minutes" — largely determine patient outcomes. Timely intervention during this window can be the difference between full recovery, long-term disability, and death.

4.2 Social Impact

Beyond individual survival, emergency response delays affect entire families and communities. A single preventable death or disability due to delayed response can push a household into long-term financial and emotional hardship, particularly in a country where informal safety nets are limited.

4.3 Economic Impact

Road traffic injuries, workplace accidents, and delayed medical response impose direct and indirect costs on the national economy — lost productivity, prolonged hospitalization, and long-term disability support. An efficient emergency coordination system reduces these downstream costs by improving first-response outcomes.



	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Number of Accidents	2636	2029	2027	2394	4312	4989	5514	5516	4198	5371
Killed	2538	1957	2067	2376	6055	7397	7221	7855	5489	6284
Injured	2134	1396	1535	1958	9914	16193	15446	13330	8600	7468

4.4 Public Safety

A population that trusts its emergency response ecosystem behaves differently — more willing to intervene, more confident in reporting emergencies, and more engaged in community safety. ResQ Plus aims to rebuild that trust through visible, structured coordination.

4.5 Healthcare System Burden

Hospitals in Bangladesh, particularly in urban centers, are frequently overloaded. Without a system to route patients efficiently based on real-time capacity and specialization, emergency cases are often sent to already-overwhelmed facilities, worsening outcomes for all patients involved, not just the one being transported.

5. Existing Situation

Today, when an emergency occurs in Bangladesh, the response process is almost entirely manual, sequential, and dependent on individual initiative rather than system design.

5.1 Typical Emergency Flow Today

1. A bystander or victim realizes help is needed.
2. They attempt to recall or search for a relevant phone number — police, fire service, a known hospital, or a personal contact.
3. Multiple calls may be needed before reaching someone who can act.
4. If an ambulance is arranged, its arrival time and route are not visible to the caller.
5. The hospital destination is often chosen based on familiarity rather than current capacity or suitability.
6. Family members are contacted informally, if at all, often well after the fact.
7. Nearby volunteers who could have helped sooner are never made aware of the incident.

5.2 Key Pain Points

- No shared visibility between citizens, hospitals, ambulances, and volunteers.
- Decision-making under stress with no guided support.
- Manual, voice-only communication that does not scale under high-demand conditions (e.g., mass casualty events, natural disasters).
- No historical or contextual data used to inform response — every emergency is handled as if it were the first.

5.3 Limitations of the Current Approach

The existing model places the entire coordination burden on the person experiencing or witnessing the emergency — precisely the person least equipped, in that moment, to coordinate anything. It is a system built around individual effort rather than structured response, which is why outcomes vary so widely from one incident to another.

6. Gap Analysis

Several isolated tools exist in the broader emergency-response space — SOS buttons, hospital directories, ride-hailing-style ambulance apps — but none of them address the underlying issue: the absence of a coordination layer connecting all emergency stakeholders in real time.

6.1 Where Current Systems Fail

Gap	Description
Fragmented Tools	Existing apps typically solve one piece (e.g., calling an ambulance) but do not connect hospitals, volunteers, and families into the same flow.
No Intelligent Triage	Most solutions do not assess the emergency type or severity — the user is expected to know what to do next.
No Real-Time Hospital Data	Bed availability, specialization, and current load are rarely visible to the public in real time.
No Volunteer Activation	Nearby capable individuals are not notified or organized during an active emergency.
No Family Loop	Emergency contacts are informed manually, late, or not at all.
No Language / Accessibility Layer	Many tools assume literacy or English fluency, excluding a large portion of the population during a crisis.

6.2 Where Communication Breaks Down

The core communication breakdown happens at the handoff points — between the citizen and the ambulance service, between the ambulance and the hospital, and between the incident scene and the family. Each handoff today depends on a phone call being correctly made, answered, and acted upon. ResQ Plus is designed specifically to replace these fragile handoffs with structured, simultaneous, system-driven coordination.

6.3 Why Coordination Is Missing

Coordination is missing because no existing stakeholder — hospitals, ambulance operators, or volunteer networks — has an incentive or platform to connect with the others. Each operates within its own operational silo. ResQ Plus is positioned as the neutral, intelligent layer that sits above these silos and connects them without requiring any one party to change how they fundamentally operate.

7. Proposed Solution

ResQ Plus is proposed as an AI-powered emergency coordination platform — a system whose primary value lies not in any single feature, but in the intelligence that connects citizens, volunteers, hospitals, ambulances, and emergency contacts into one coherent response during a crisis.

7.1 The Philosophy

The platform is built on a simple principle: during an emergency, the person in distress should not be responsible for coordinating their own rescue. ResQ Plus takes on that coordination role — assessing the situation, guiding immediate action, and activating the right people and resources in parallel rather than in sequence.

7.2 Why a Coordination Platform, Not Just an App

A conventional SOS application typically performs one function — sending an alert. ResQ Plus is architected differently: it treats every emergency as a multi-party event requiring simultaneous action from several stakeholders. This distinction is central to the product's identity.

> *"ResQ Plus doesn't replace emergency services — it intelligently connects them."*

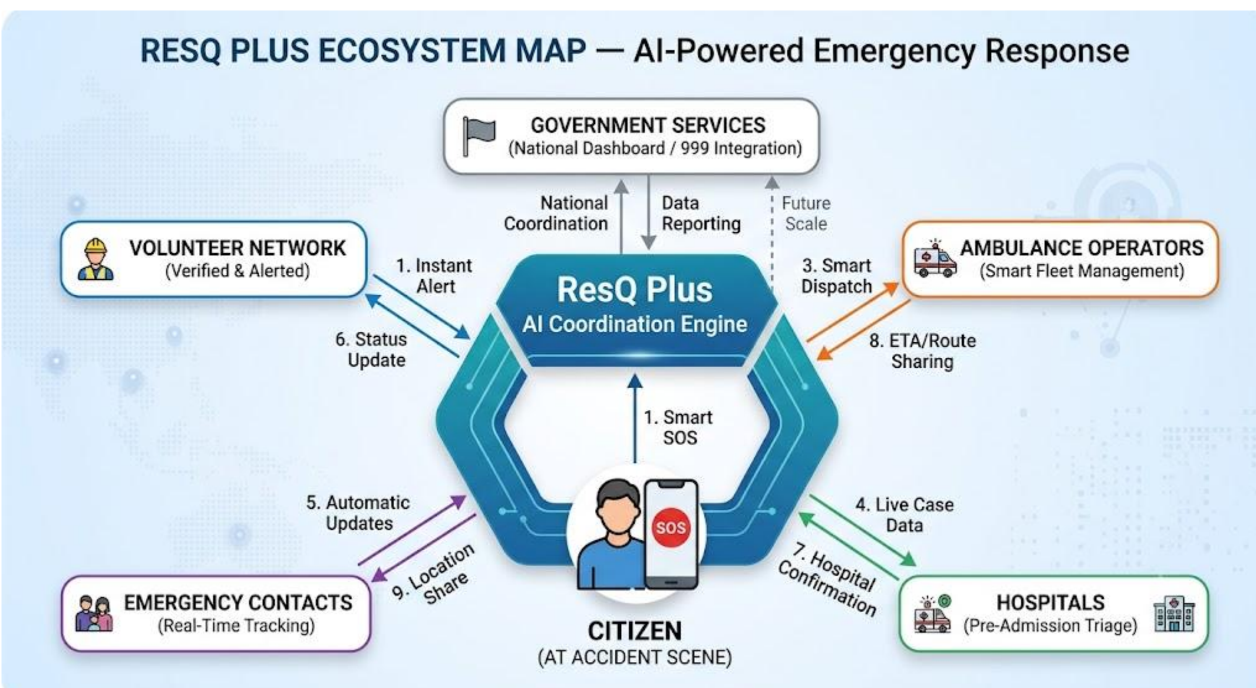
Conventional SOS App	ResQ Plus
Sends a single alert to one contact	Coordinates multiple stakeholders simultaneously
No guidance during the emergency	AI-guided first-response assistance
No hospital intelligence	Recommends appropriate, nearby, capacity-aware hospitals
No volunteer network	Activates trained/nearby volunteers in real time
Static contact list	Continuous family/contact updates throughout the event
Isolated tool	Ecosystem designed for future institutional integration

8. Product Overview

ResQ Plus operates as an ecosystem rather than a single-user tool. Each stakeholder interacts with the platform through a role-specific interface, while the underlying AI coordination engine ensures all parties are working from the same real-time picture of the emergency.

8.1 The ResQ Plus Ecosystem

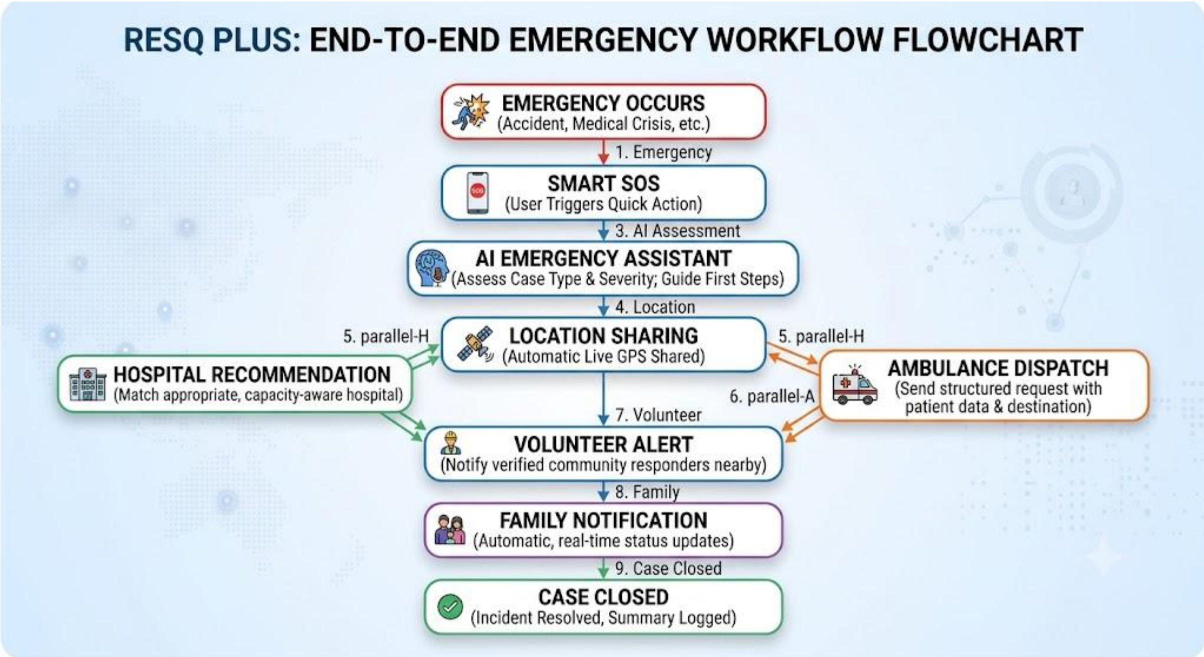
- **Citizen:** The primary user who experiences or witnesses an emergency and initiates a Smart SOS.
- **Volunteer:** Verified or community-registered individuals nearby who can be alerted and mobilized to assist before professional help arrives.
- **Hospital:** Facilities that receive real-time case information and can confirm readiness, enabling faster admission upon arrival.
- **Ambulance:** Service providers who receive structured dispatch requests with location, case type, and destination guidance.
- **Emergency Contacts:** Pre-registered family or friends who receive automatic, real-time status updates without needing to be manually contacted.
- **Future Government Integration:** A planned connection layer to national emergency services (police, fire, disaster management) for institutional-scale coordination.



9. How ResQ Plus Works

The ResQ Plus workflow is designed to compress what is normally a sequence of manual actions into a single, guided, largely automated flow.

8. Emergency Occurs — A medical, accident, or safety emergency takes place.
9. User Opens the App — The citizen (or a bystander) opens ResQ Plus, or triggers it through a pre-set quick action.
10. Smart SOS — A single action initiates the emergency flow, sharing the user's situation with the coordination engine.
11. AI Assistance — The AI assistant asks brief, guided questions to understand the nature and severity of the emergency and offers immediate first-response guidance.
12. Location Sharing — Precise, live location is shared automatically with relevant responders.
13. Volunteer Alert — Nearby registered volunteers are notified and can choose to respond.
14. Hospital Recommendation — The system suggests the nearest suitable hospital based on emergency type and available information.
15. Ambulance Dispatch — A structured request is sent to available ambulance services with case details and destination.
16. Family Notification — Pre-registered emergency contacts automatically receive real-time status updates.
17. Case Closed — Once the situation is resolved, the case is marked complete and a summary is logged for the user's records.



10. Target Users

10.1 Primary Users

- Individuals and families seeking a faster, more reliable way to respond to medical and accident emergencies.
- Commuters and road users, given the high frequency of road-traffic incidents.

10.2 Secondary Users

- Community volunteers and civic responders willing to assist during emergencies.
- Ambulance service operators seeking better dispatch efficiency.

10.3 Institutional Users

- Hospitals and clinics seeking improved incoming-patient visibility.
- Corporate offices and educational institutions adopting ResQ Plus for staff and student safety.
- NGOs and community safety organizations.

10.4 Future Users

- Government emergency and disaster management agencies.
- Insurance providers integrating emergency response data into policy services.

11. Core Features (MVP)

The Minimum Viable Product for ResQ Plus is scoped around six core features, each addressing a distinct point of failure identified in the Problem Statement.

11.1 Smart SOS

A single, prominent action that immediately initiates the emergency flow — capturing location, opening AI assistance, and preparing to alert relevant parties, without requiring the user to navigate menus under stress.

11.2 AI Emergency Assistant

A conversational assistant that asks minimal, targeted questions to understand the emergency, offers clear first-response guidance, and helps the user or bystander make informed decisions in the absence of professional help.

11.3 Live GPS Tracking

Continuous, accurate location sharing with volunteers, ambulance services, and emergency contacts, replacing verbal or text-based address descriptions that are often unclear or delayed.

11.4 Hospital Recommendation

A recommendation feature that surfaces nearby hospitals suited to the emergency type, using available facility information to reduce misdirected transport.

11.5 Volunteer Network

A structured system for registering, verifying, and alerting nearby volunteers, turning informal civic goodwill into a coordinated, trackable response resource.

11.6 Emergency Contact Notification

Automatic, real-time updates sent to pre-registered emergency contacts throughout the incident, removing the burden of manual notification from the person handling the crisis.

12. AI Integration

Artificial Intelligence is the coordination engine of ResQ Plus — not a marketing feature, but the mechanism that makes real-time, multi-party response possible. The following describes how AI is realistically applied within the MVP and near-term roadmap.

12.1 Emergency Assessment

AI-guided questioning helps classify the type and likely severity of an emergency based on the user's responses, enabling more relevant next steps than a generic alert.

12.2 Decision Support

The assistant provides clear, step-by-step first-response guidance appropriate to the situation, helping bystanders act correctly instead of freezing or improvising under pressure.

12.3 Voice Interaction

Voice-based interaction allows users to engage with the assistant hands-free — an important consideration when a user may be injured, driving, or physically assisting a victim.

12.4 Priority Prediction

Based on reported symptoms or incident type, the system can assist in estimating relative urgency, helping responders and dispatchers prioritize when multiple cases occur simultaneously.

12.5 Hospital Recommendation

AI matches emergency type with facility suitability and proximity to suggest an appropriate destination, rather than defaulting to the nearest or most familiar hospital regardless of fit.

12.6 Multilingual Assistance

Given Bangladesh's linguistic landscape, the assistant is designed to operate in Bangla and English, with regional language support considered for future phases.

12.7 Future AI Possibilities

- Predictive analytics for high-risk zones and peak incident times.
- Automated severity triage integrated directly with hospital systems.
- Pattern recognition for disaster-scale, multi-casualty coordination.

> **A Note on Realism**
>
> *ResQ Plus does not claim AI can replace medical professionals or emergency responders. Its role is strictly to assist, guide, and coordinate — compressing decision time and improving the quality of information available to human responders.*

13. Unique Selling Proposition

The emergency-tech space is not short of SOS buttons. It is short of coordination. ResQ Plus's core differentiation is not a single feature — it is the decision to build a coordination layer rather than another isolated alert tool.

> *"ResQ Plus doesn't replace emergency services — it intelligently connects them."*

13.1 Why Coordination Is More Valuable Than Another SOS App

- An SOS app that only sends an alert leaves every subsequent step — dispatch, hospital choice, family notice — exactly as fragmented as before.
- ResQ Plus removes the fragmentation itself by acting on multiple fronts simultaneously, the moment an emergency is reported.
- This positions ResQ Plus not as a competitor to any single existing service, but as the connective layer that makes all of them work better together.

13.2 Defensibility

Because ResQ Plus's value comes from network effects — more hospitals, volunteers, and users making the system smarter and faster for everyone — it becomes structurally difficult for a single-feature competitor to replicate without building the same multi-stakeholder ecosystem.

14. Business Model

ResQ Plus is designed with a multi-stakeholder monetization approach, reflecting the reality that emergency coordination creates value for several different parties — not just the end citizen.

Stakeholder	Potential Revenue / Partnership Model
Government	Public-sector partnership or licensing for national emergency coordination infrastructure.
Hospitals	Subscription access to incoming-patient visibility and case coordination tools.
NGOs	Partnership and sponsorship for volunteer network expansion and community outreach.
Corporate Safety	B2B subscription for employee safety coverage in offices, factories, and campuses.
Insurance Companies	Data-sharing and referral partnerships tied to emergency response records.
Premium Services	Optional premium tier for individuals/families — e.g., priority routing, extended contact lists, family tracking.
Future API Integration	API licensing for third-party platforms (ride-hailing, smart city systems) to plug into ResQ Plus coordination.

The core citizen-facing emergency features are intended to remain free or low-cost at all times, given the public-safety nature of the platform; monetization is designed to come primarily from institutional and premium-tier stakeholders rather than from restricting access during an emergency.

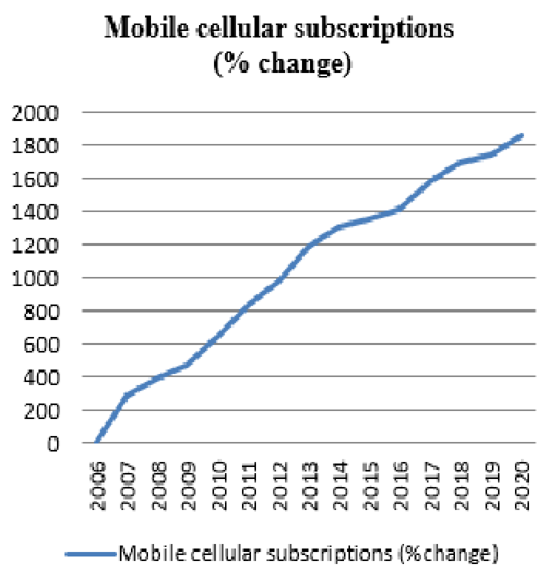
15. Market Opportunity

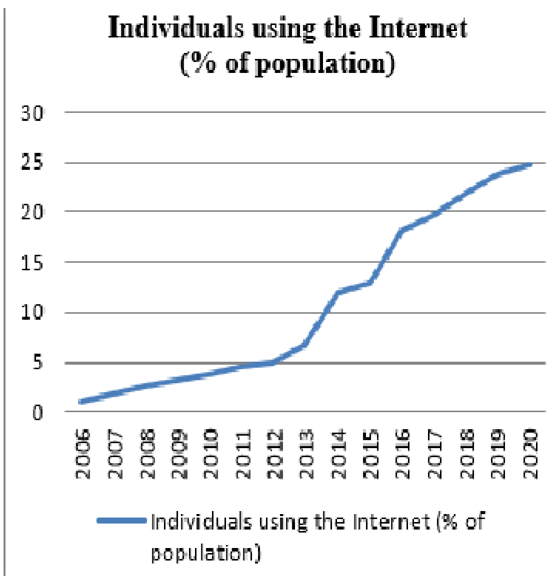
15.1 A Country in Digital Transition

Bangladesh is undergoing rapid digital transformation, driven in part by the national "Smart Bangladesh" vision, which prioritizes digital infrastructure, e-governance, and technology-enabled public services. ResQ Plus is directly aligned with this direction — applying digital and AI infrastructure to one of the most tangible public-good problems: emergency response.

15.2 Smartphone and Internet Adoption

Smartphone penetration and mobile internet usage in Bangladesh continue to grow steadily, particularly among the urban and semi-urban population most exposed to road-traffic and workplace emergencies.





This growing base of connected users represents the addressable market for a mobile-first coordination platform.

15.3 Why Now

- Government digital initiatives create favorable conditions for public-private emergency-tech partnerships.
- Rising smartphone and mobile data adoption removes a key historical barrier to app-based emergency response.
- Increasing urban traffic density and associated road accident rates raise the urgency of faster response systems.
- No existing player has captured the coordination layer described in this proposal, leaving room for a first-mover advantage.

16. Implementation Roadmap

ResQ Plus is proposed to move through four structured phases, each building the credibility, data, and partnerships required for the next.

Phase 1 — Prototype

- Develop core MVP: Smart SOS, AI Assistant, GPS tracking, and contact notification.
- Conduct internal testing and usability refinement.
- Build initial partnerships with a small number of hospitals and volunteer groups for validation.

Phase 2 — Pilot

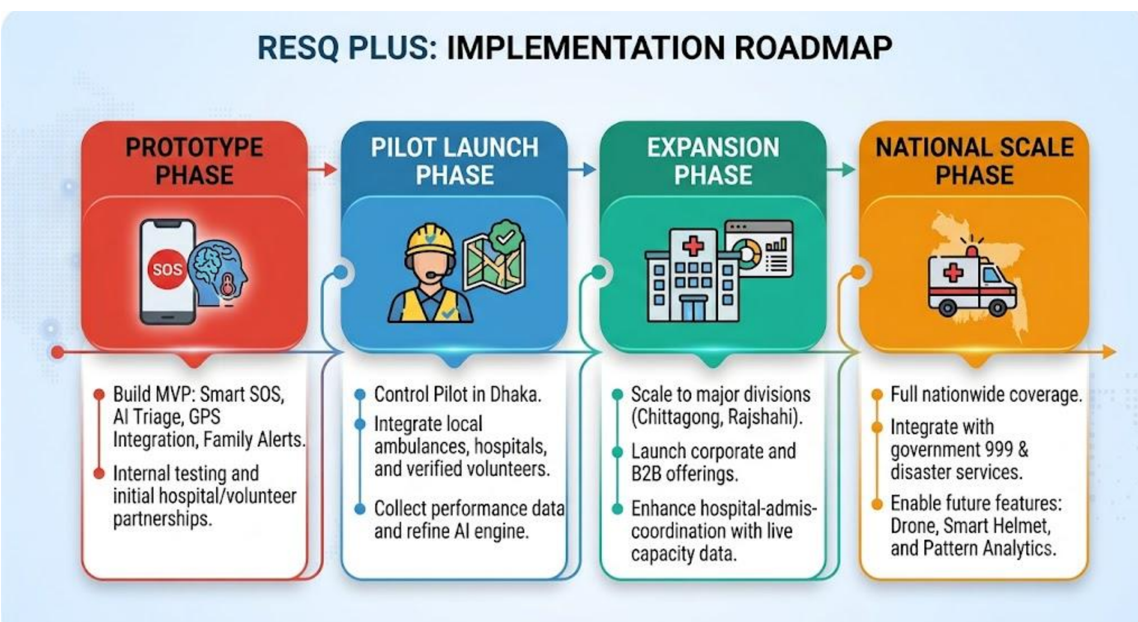
- Launch a controlled pilot in one city or district.
- Onboard a defined set of hospitals, ambulance operators, and verified volunteers.
- Collect real-world performance data and refine the AI assistant and hospital recommendation logic.

Phase 3 — Expansion

- Expand to additional cities and divisions based on pilot learnings.
- Broaden hospital and ambulance partnerships.
- Introduce premium and institutional (corporate/NGO) offerings.

Phase 4 — National Scale

- Pursue integration with national emergency services and government disaster management systems.
- Scale the volunteer network nationwide.
- Explore regional expansion beyond Bangladesh where similar coordination gaps exist.



17. Expected Impact

Impact Area	Expected Outcome
Lives Saved	Faster, guided response during the golden minutes improves survival and recovery outcomes.
Faster Response	Simultaneous coordination replaces slow, sequential phone-based communication.
Better Coordination	Citizens, hospitals, ambulances, and volunteers operate from a shared real-time picture.
Healthcare Efficiency	Appropriate hospital routing reduces misdirected patients and unnecessary transfers.
Community Engagement	Structured volunteer activation strengthens civic participation in public safety.
Volunteer Empowerment	Everyday citizens are given a clear, trackable way to help during emergencies.

18. Future Roadmap

Beyond national scale, ResQ Plus is designed to evolve into a broader public-safety intelligence platform.

- **Wearable Integration:** Connecting with smartwatches and health wearables for automatic vital-sign sharing during emergencies.
- **Crash Detection:** Automatic accident detection using device sensors, triggering SOS without manual action when a user is incapacitated.
- **Drone Assistance:** Exploring drone-based first-look assessment or medical-supply delivery in hard-to-reach areas.
- **Disaster Response:** Extending the coordination model to floods, fires, and other large-scale disaster scenarios.
- **Government Dashboard:** A centralized command view for authorities to monitor and coordinate citywide or national emergency activity.
- **Analytics:** Aggregated, anonymized insights on incident hotspots and response performance to guide public safety policy.
- **IoT Integration:** Connecting with smart-city infrastructure such as traffic sensors and public safety cameras for faster incident detection.

19. Conclusion

Emergencies will always be unpredictable. What should not be unpredictable is whether help arrives in time, whether it arrives at the right place, and whether the people who matter most are kept informed along the way. That is the problem ResQ Plus exists to solve.

ResQ Plus is not proposed as another application competing for space on a phone's home screen. It is proposed as the coordination layer that Bangladesh's emergency response ecosystem currently lacks — one that respects the roles of hospitals, ambulance services, and emergency authorities, and makes each of them faster and more effective by connecting them intelligently.

Built with realism, guided by AI where it genuinely adds value, and designed to scale from a single pilot city to a national infrastructure layer, ResQ Plus represents a practical, high-impact opportunity to save lives through better coordination — not more complexity.

> *“ResQ Plus doesn’t replace emergency services — it intelligently connects them.”*